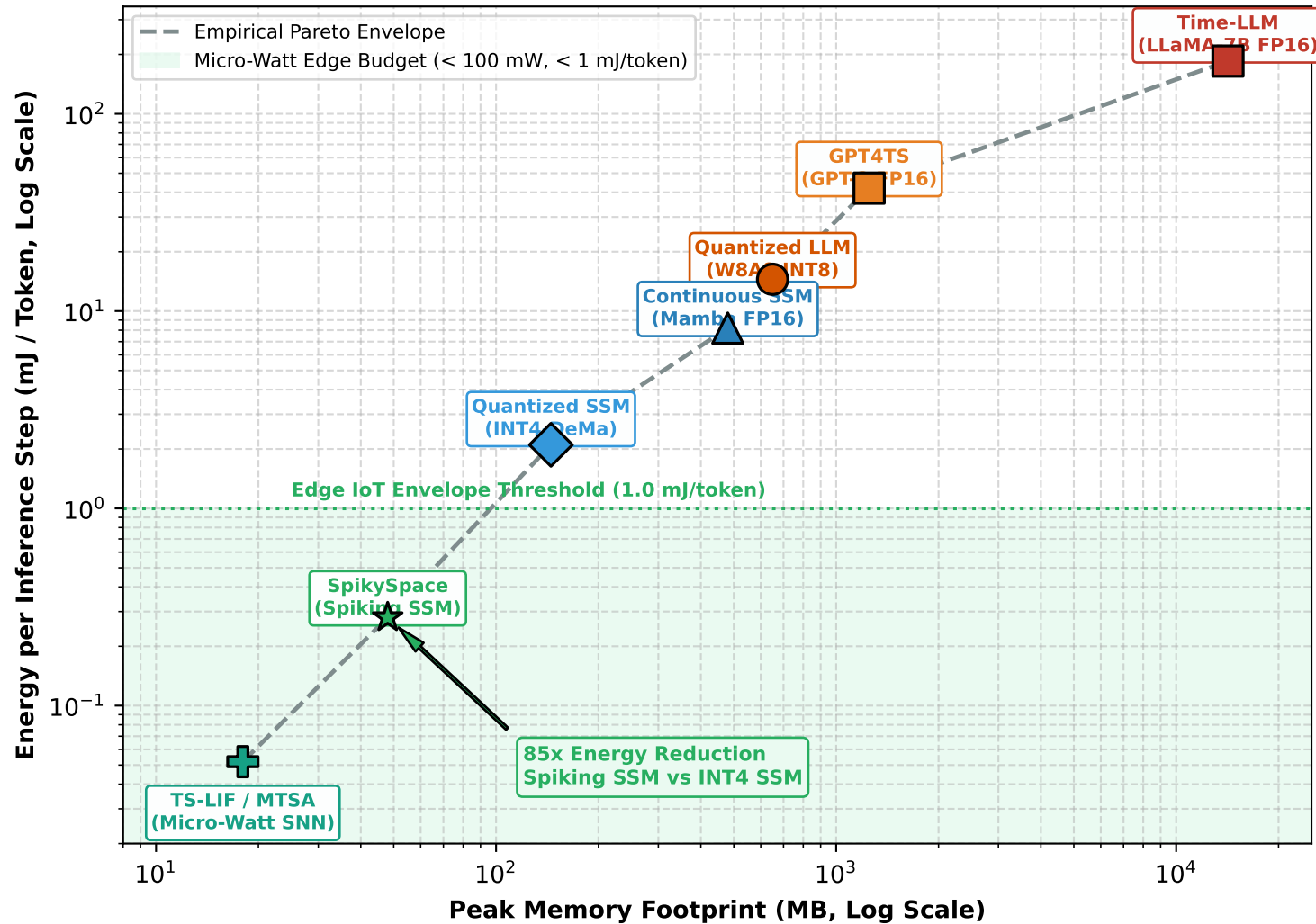


**(a) Energy Efficiency vs. Memory Footprint Pareto Frontier**



**(b) Dual-Compartment Spiking Dynamics (TS-LIF / MTSA-SNN)**

